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# QUOTATION

62.4 kWp on-grid solar PV system with zero-export protection

| | | | |
|---|---|---|---|
| **Customer** | Sample: customer details removed | **Date** | 22 October 2026 |
| **Site** | Factory, northern Thailand | **Validity** | 30 days (sample prices) |
| **System** | 96 panels (6 strings × 16 panels), 50 kW inverter, metal-sheet roof mount | **Currency** | Thai baht (THB) |

**System overview.** 96 LONGi Hi-MO X10 Explorer 650 W monofacial panels (62.4 kWp) connected to a Huawei SUN2000-50K-MC0 inverter in a 1/2/2/1 string layout across MPPT 1–4, the most efficient arrangement for this inverter. Operating voltage is about 603 V. Expected generation is about **89,300 kWh per year**. Includes a zero-export meter and monitoring through the FusionSolar app.

| No. | Description | Unit | Qty | Unit price | Amount (THB) |
|---|---|---|---|---|---|
| **A. Solar panels and inverter** | | | | | |
| 1 | **Solar panels**<br>LONGi Hi-MO X10 Explorer LR7-72HVH-650M 650 W monofacial, 2382 × 1134 × 30 mm | panel | 96 | 3,276.00 | 314,496.00 |
| 2 | **Inverter**<br>Huawei SUN2000-50K-MC0 50 kW 3-phase, 4 MPPT / 8 inputs, with AFCI · 5-year warranty | set | 1 | 89,917.30 | 89,917.30 |
| 3 | **Zero-export meter**<br>Huawei DTSU666-H 3-phase, CTs included | set | 1 | 2,904.68 | 2,904.68 |
| 4 | **Monitoring device**<br>Huawei SDongleB-06 4G + FusionSolar app | set | 1 | 5,004.67 | 5,004.67 |
| **B. Mounting system for metal-sheet roof** | | | | | |
| 5 | **Mounting rail**<br>4.8 m aluminium rail; 6 rows of 16 panels, 2 rails per row | length | 48 | 516.17 | 24,776.16 |
| 6 | **Rail splice**<br>Joins rails into a continuous run along each row | pc | 36 | 35.32 | 1,271.52 |
| 7 | **Metal-roof L-feet**<br>L-brackets fixed to the roof purlins at max. 1.2 m spacing, with screws and EPDM seals (incl. 5% spare) | pc | 214 | 34.35 | 7,350.90 |
| 8 | **Rail T-bolt M8**<br>Fixes the L-feet to the rails | set | 214 | 9.14 | 1,955.96 |
| 9 | **Mid clamp**<br>For 30 mm panel frames, stainless-steel bolts (incl. 5% spare) | pc | 189 | 16.68 | 3,152.52 |
| 10 | **End clamp**<br>For 30 mm panel frames, stainless-steel bolts (incl. spare) | pc | 26 | 16.68 | 433.68 |
| 11 | **Grounding clip**<br>Bonds each panel frame to the rail | pc | 192 | 5.04 | 967.68 |
| 12 | **Roof waterproofing**<br>PU sealant and butyl tape for every fixing point | lot | 1 | 3,412.50 | 3,412.50 |
| **C. DC electrical system** | | | | | |

| No. | Description | Unit | Qty | Unit price | Amount (THB) |
|---|---|---|---|---|---|
| 13 | **PV cable, red**<br>LINK H1Z2Z2-K 1×6 mm² 1.5 kV DC, 100 m roll, EN 50618 | roll | 3 | 3,071.51 | 9,214.53 |
| 14 | **PV cable, black**<br>LINK H1Z2Z2-K 1×6 mm² 1.5 kV DC, 100 m roll, EN 50618 | roll | 3 | 3,071.51 | 9,214.53 |
| 15 | **PV connectors**<br>MC4 male/female pairs, 1500 V (incl. spare) | pair | 18 | 28.47 | 512.46 |
| 16 | **DC combiner box**<br>Waterproof steel enclosure KJL KBSN 04 | set | 1 | 1,251.17 | 1,251.17 |
| 17 | **DC breaker**<br>Suntree SL7N-63 4P 32 A 1000 V DC, one per string | pc | 6 | 676.13 | 4,056.78 |
| 18 | **DC surge protection (SPD)**<br>BENY BUD40/3 Type II 3P 1000 V DC, one per MPPT | pc | 4 | 657.30 | 2,629.20 |
| **D. AC electrical system** | | | | | |
| 19 | **AC enclosure**<br>Steel enclosure KJL KBSS9004 (44 × 61 × 23 cm) | set | 1 | 936.17 | 936.17 |
| 20 | **Inverter output breaker**<br>CHINT NXM-125S MCCB 3P 100 A 25 kA | pc | 1 | 936.39 | 936.39 |
| 21 | **AC surge protection (SPD)**<br>Suntree SUP1-40 Type II 4P 40 kA 385 V AC | pc | 1 | 500.47 | 500.47 |
| 22 | **AC cable (phases and neutral)**<br>THW 25 mm² (4 cores × 30 m) | m | 120 | 156.14 | 18,736.80 |
| 23 | **AC earth cable**<br>THW 16 mm² green/yellow | m | 30 | 99.24 | 2,977.20 |
| 24 | **Conduit and fittings**<br>IMC/EMT 1¼" conduit, boxes, clamps and glands | lot | 1 | 16,380.00 | 16,380.00 |
| **E. Earthing system** | | | | | |
| 25 | **Earth rods**<br>5/8" × 3 m copper rods with clamps and inspection pits | pc | 3 | 649.80 | 1,949.40 |
| 26 | **Main earth conductor**<br>16 mm² green/yellow to the main earth bar | m | 50 | 99.24 | 4,962.00 |
| 27 | **Equipotential bonding**<br>6 mm² green/yellow for structure, rails and enclosures | m | 100 | 38.00 | 3,800.00 |
| **F. Services** | | | | | |
| 28 | **Installation**<br>Mounting, panels, inverter, electrical work and labelling | lot | 1 | 131,040.00 | 131,040.00 |
| 29 | **Roof-work safety equipment**<br>Lifeline, roof walkways and fall protection during installation | lot | 1 | 10,500.00 | 10,500.00 |
| 30 | **Testing and commissioning**<br>Tests to IEC 62446: polarity, Voc, Isc, insulation resistance, earth resistance and zero-export check | lot | 1 | 15,750.00 | 15,750.00 |
| 31 | **PEA documents**<br>PEA grid-connection application with SLD certified by an electrical engineer | lot | 1 | 15,750.00 | 15,750.00 |

| No. | Description | Unit | Qty | Unit price | Amount (THB) |
|---|---|---|---|---|---|
| 32 | **Transport**<br>Delivery of equipment to site | lot | 1 | 10,500.00 | 10,500.00 |

| | |
|---|---|
| **Equipment subtotal (sections A–E)** | **533,700.67** |
| **Services subtotal (section F)** | **183,540.00** |
| **Total before VAT** | **717,240.67** |
| VAT 7% | 50,206.85 |
| **Grand total** | **767,447.52** |

Equipment prices exclude delivery; transport is shown separately as item 32. Services in section F are subject to 3% withholding tax (5,506.20 THB), which the customer deducts at payment and certifies to the company with a withholding-tax certificate.

## Optional items (not included in the total above)

| No. | Description | Unit | Qty | Unit price | Amount (THB) |
|---|---|---|---|---|---|
| O1 | **Soft starter or VFD (optional)**<br>22 kW 3-phase for the 18 kW compressor motor, with bypass contactor | set | 1 | 34,125.00 | 34,125.00 |
| O2 | **Ice-machine circuit breaker (optional)**<br>MCCB 3P 63 A in the MDB | pc | 1 | 3,822.00 | 3,822.00 |

## Loads the solar system will supply (site photos)



Photo 1 — Tube-ice machine, the main load of the system

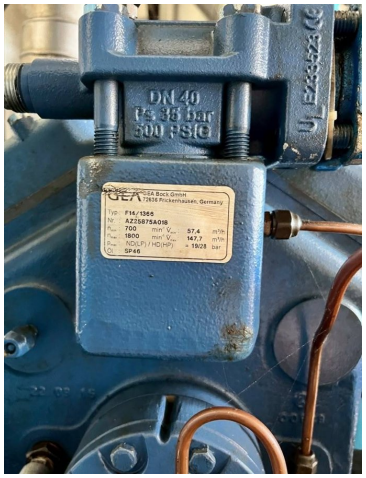



Photo 2 — GEA Bock F14/1366 compressor, swept volume 57.4–147.7 m³/h at 700–1,800 rpm, max. pressure 19/28 bar




Photo 3 — Müller Motor 3-phase induction motor of the auxiliary equipment

The 62.4 kWp system is designed to supply the ice machine and compressor (combined load about 20–22 kW) during the day. Any demand above the solar output is drawn from the PEA grid as normal, and the zero-export meter stops power flowing back into the grid. A soft starter or VFD (item O1) is recommended to reduce inrush current when the compressor starts.

## Payment terms

| No. | Condition | Share | Amount (THB) |
|---|---|---|---|
| 1 | On signing the contract | 40% | 306,979.01 |
| 2 | When the main equipment (panels, inverter, mounting) is delivered to site | 50% | 383,723.76 |
| 3 | After testing and commissioning are complete | 10% | 76,744.75 |

## General conditions

1. Work period is 45 working days from contract signing and receipt of the first instalment in full (excluding rain days and holidays). The customer prepares the site ready for work to start.
2. Warranty: LONGi panels 15-year product and 30-year performance warranty · Huawei inverter 5 years · installation workmanship 2 years, under the manufacturers' and the company's conditions.
3. Includes the PEA grid-connection application with an SLD certified by an electrical engineer. PEA fees (if any) are not included.
4. Installed on the metal-sheet roof with rails, L-feet and panel clamps; system weight about 13–15 kg/m². The company inspects the purlins and roof structure before installation. If the roof structure needs strengthening or existing roof sheets are damaged, this is quoted separately. Every fixing point is sealed with rubber seals and sealant.
5. Upgrades to the existing MDB are not included. If the site survey shows they are needed, they are quoted separately.
6. Monitoring through the FusionSolar app with automatic alerts. 24-hour monitoring service is not included.
7. Copper cables and certified protection devices are used, and the work is carried out by certified electricians.
8. Equipment prices exclude delivery; delivery to site is shown separately as item 32.

This is a sample quotation published for reference. Customer details have been removed. Prices are indicative for this design as of October 2026; ask Solar ESS for a quotation for your own site.